Amit Kumar Singh

thegigasurgeon@gmail.com · +91-9915503368 · gigasurgeon.github.io/portfolio ·
www.linkedin.com/in/thegigasurgeon

Experience

Real Brokerage — Delhi, India
AI ML Engineer — Oct 2024 - Present

- Pretrained and Finetuned of state-of-the-art LLMs and VLMs with reasoning and thinking capabilities, including **LLaMA 3.3, Qwen 2.5 72B, Qwen 3, Gemma**, using **LoRA/QLoRA and GRPO-based RL post-training**, tailored for real estate auto-review.
- Currently leading development of enterprise-scale agentic AI document intelligence platform processing **80K+ PDFs daily**, leveraging **multi-agent text+visual reasoning pipelines** including **tool-calling, subagents, skills.**
- Built a scalable **Multi-LLM Orchestration** Infrastructure to facilitate a lot of automation within the organization saving **1000s** of developer hours every week.
- Finetuned **Google Gemini-3.5-Pro, Gemini-3.5-Flash, GPT 4o** for robust multi-llm agentic workflow.
- End-to-End precise RAG pipeline using **intent routing, semantic chunking**, **llm reasoning**, **hybrid vector retrieval**, **context optimization**, **multi-source context aggregation**, embedding generation and structured output generation.
- **Claude Agent SDK** integration for automated PR reviews, incident investigation, and webhook-driven workflows
- Led an 8-person developer's team coordinating validator development across jurisdictions.
- Built high-precision information extraction pipelines for unstructured real estate documents using **YOLOv9, Doctr OCR, and custom VLMs**, boosting accuracy from **35% to 99.85%**.
- Engineered scalable ML pipelines fusing **5+ specialized models** (VLMs, LLMs, OCR, Object Detection, embeddings) in a single application, deployed as **microservices on Kubernetes** across AWS and GCP, processing **80k PDFs** per day.
- Achieved a **200%** pipeline speedup through graph optimizations, batched inference, model quantization (**INT8/FP16, AWQ/GPTQ**), and architecture-level modifications for fast inference.

Streamingo AI — Noida, India
Deep Learning Engineer — Nov 2022 – Oct 2024

- Developed end-to-end temporal event understanding systems using **Swin, VideoMAE, MVD, ViT, BEiT**.
- Worked on problems like video activity recognition, player tracking, pose detection, basketball court localization, face blurring, logo identification, zero shot classification, planogram compliance, etc. on videos and images.
- Designed zero-shot and few-shot classification systems using **CLIP, Detic, Prototypical Networks**, enabling rapid scalability across unseen classes.
- Built a real-time sports analytics engine combining object detection, multi-object tracking, pose estimation, and court localization.
- Deployed optimized inference pipelines using **TensorRT** and ONNX, delivering **3x** FPS.

Attentive AI — Noida, India
Computer Vision Research Engineer — Apr 2021 – Nov 2022

- Built and trained State Of The Art neural networks for image segmentation in satellite imagery i.e. **HRNet, Vision Transformer**.
- End to end ownership of CV products (Falcon and Respod) across the company.
- Improved accuracy of models by 12% and reduced overall post-processing time by 90%.

Nokia — Bangalore, India
Software Engineer — Jul 2020 – Mar 2021

- Created a meta-learning platform for dynamic model selection and hyperparameter optimization across various business problems.

- Integrated Explainable AI (XAI) modules using SHAP, LIME for model interpretability and compliance.

Projects

Basketball Video to 3D Mapping *(PyTorch, Visual SLAM, ORBv3, Blender, Homography, VideoMAE)*
Reconstructed player trajectories in 3D from monocular sports footage using deep video transformers and traditional SLAM pipelines. Added temporal event summarization via transformer-based description generation.

Tekken with Pose *(YOLOvNAS, Pose Estimation, Raspberry Pi)*
Created a camera-driven interface to map real-world actions into virtual gameplay using pose estimation, latency-aware gesture recognition, and control mapping.

Charlie - AI Virtual Assistant *(Raspberry Pi, Python, STT, FaceNet)*
Built an edge-deployable assistant with multi-modal recognition: speaker ID, facial login, offline STT, and contextual response generation.

Skills

Programming Languages:	Python, C++, JavaScript
ML / Post-Training:	GRPO (RL), LoRA/QLoRA, distillation, quantization (INT8/FP16, AWQ/GPTQ), agentic AI (tool-calling, subagents), RAG, multimodal fine-tuning
Libraries & Infra:	PyTorch, TensorFlow, Langchain, Claude Agent SDK, Detectron2, OpenCV, TensorRT, ONNX, Kubernetes, Docker, microservices, AWS, GCP

Education

Chandigarh University B.Tech. in Computer Science
GPA: 8.28